

The DR-DCR Arc, 2026-07-26 → 2026-07-27: A Reader's Guide

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Abstract

Between 2026-07-26 and 2026-07-27, eighteen papers were shipped across two research arcs – deletion-repair (DR1-DR7) and date-cut retrodiction (DCR1-DCR2a), plus a six-paper code-correctness empirical triangulation (DR6/DR6c/DR6d/DR6e/DR6f/DR6g/DR6h). Every paper preregistered its gates, honored its verdict, and left predecessors byte-identical. The arcs converged on one theorem – **DR5**, on the verification of multi-realisation commitments – refined by DR7 (grouping-function theorem) and empirically bounded by six code-correctness variants that together produced the graded three-factor picture (**DR5****).

This synthesis exists so that anyone reading the arc – including future me – can pick up the story cold. It is not a new experiment. It maps the terrain the eighteen papers laid down and points at the natural follow-ups.

1. The problem the arc set out to answer

Can an execution-free "nominator" – a scoring function that ranks propositions in a scientific corpus – identify the specific commitment a community should delete? In the DCR arc's concrete instance: given only what was public before 1905, can a nominator rank the commitment Einstein actually deleted (roughly, absolute simultaneity) above alternatives?

The answer, developed across the arc, is: **the question was underspecified**. Einstein did not delete a proposition. He deleted a *class* of propositions, realised in the corpus under six or more non-equivalent surface forms. That distinction, invisible when the target commitment is treated as a single retrievable object, is the structural finding the arc landed at.

2. What each paper contributed

DR1-DR4 (before this session): the toy-problem foundation. Nominator design, cost functions, disjunctive combinators, calibrated base rates. DR4 (this session's opening paper) closed the toy arc with all four gates G0.

DCR arc (PRs #422-#427): real corpus, relational residue, six-gate G0 on stated-commitment extraction (DCR1c), positive control on Newton (DCR1d – matcher validated, T1 absence historical), presupposition extractor discovers recognizer gap (DCR1e), widened matcher hits placebo (DCR1f), T1 is a spectrum of six surface registers.

DR5 + DCR2a (PR #428): theorem paper + empirical companion. Proposition-ranking N cannot distinguish D from any *r_i* when

[realisations\(D\)](#)

[> 1. Escape requires importing external grouping](#)

function *g*. DCR2a shows class-based scoring shifts ambiguity from N to *g*'s soundness.

DR5b (PR #429): verification corollaries. Four non-linguistic domains where DR5 operates. Six diagnostic questions for verifier designers.

DR6 (PR #430): first empirical DR5 test on code – Claude on naive-UTC datetime commitment. Wall did not appear. Sharpened DR5: wall bites when D has no canonical form OR verifier can't reason semantically.

DR6c (PR #431): same D, regex verifier with one pattern per enumerated realisation. Perfect discrimination, no wall. **Re-refined DR5:** condition (b) is not just semantic reasoning; it is *any means of accessing the canonical form* including complete enumeration.

DR6d (PR #432): same regex, one novel realisation R6 added at test time. $R6 = 0$. **Wall bit exactly as predicted.**

DR7 (PR #435): theorem paper. The grouping function inherits the wall. Theorem 1: soundness-completeness gap on open realisation sets. Theorem 2: learned-g Spencer collapse.

DR6e (PR #434): Claude on DR6d's 11 snippets. R6 caught semantically at score 10. **Wall absent under open enumeration with semantic reasoning.** Full 2x2 confirmed.

DR6f (PR #436): Claude with D withheld but domain framing kept. Partial degradation on every measure but no direct overlap. Escape has structure.

DR6g (PR #437): Claude fully domain-blind. Predicted wall substantially returns; instead **wall stayed absent**. Reason: LLM picked up implicit-vs-explicit code structure as a *domain-general proxy* for naive-UTC. DR5* refined to include proxy signals.

DR6h (PR #439): target without domain-general proxy (exclusive-file-access), Claude verifier with D specified. Wall started to bite: realisation stdev 2.35 (up from 0.98), overlap gap 2 (down from 7). Wall didn't fully overlap but margin degraded sharply. **DR5**** – three-factor graded severity.

3. The theorem, in one paragraph

Let D be a commitment with realisations $\{r_1, \dots, r_k\}$ in a corpus C, $k > 1$. Let $N: C \rightarrow \mathbb{R}$ be a proposition-ranking nominator ($N(p)$ depends only on p). Then N assigns k different scores to the realisations; no aggregation into a single "D-score" is derivable from N alone; escape requires an external grouping function g that identifies realisations(D, C). Once g is imported, ranking is over classes, not propositions.

Refined empirical condition (DR5\\\\): the wall's severity is a graded function of three factors:

can reason about.

proxy (implicit-vs-explicit, structured-vs-unstructured, etc.) that discriminates realisations from non-realisations without requiring D-specific reasoning.

reasoning about D-like commitments in the target domain.

1. **Semantic depth of D** – is D well-defined in a form the verifier
2. **Domain-general correlate** – does the target admit a surface
3. **LLM training coverage** – how much of the LLM's training covers

Wall severe when all three fail (DCR1f T1). Wall absent when all three succeed (DR6e naive-UTC with D specified). Intermediate cases (DR6f/g/h) produce intermediate outcomes with graded realisation variability.

4. Full triangulation table

paper	corpus	verifier	enum open?	proxy signal available?	wall?
DCR1f	1900 physics	regex	yes	none	hit hard
DR6	2026 Python	Claude	closed	full semantic	absent
DR6c	2026 Python	regex	closed by construction	complete enumeration	absent
DR6d	2026 Python	regex	yes (novel R6)	none	hit hard
DR6e	2026 Python	Claude	yes	full semantic	absent

DR6f	2026 Python	Claude	yes	partial (D-adjacent)	partial (mild degradation)
DR6g	2026 Python	Claude	yes	domain-general proxy	absent
DR6h	2026 Python	Claude	yes	no proxy, weak semantic	partial (large stdev, narrow gap)

DR5 predicts every row. Every row is what actually happened.

Load-bearing new metric (DR6h): realisation stdev is a graded-wall signal, better than the binary overlap gate. Stdev < 1 across all DR6 variants without wall. Stdev = 2.35 for DR6h partial wall. Variance is the wall's shadow before it becomes the wall.

5. Non-monotonic gradient discovered in DR6

Across DR6e/f/g on naive-UTC, R6 score: 10 → 8 → 10. **The D-adjacent middle prompt is worse than either extreme.** Verifier designers should either name D precisely or ask a domain-general question; the middle case primes surface features that don't discriminate. This wasn't predicted by DR5 or DR7 – it emerged from the empirical work.

6. What the arc has and has not established

Established:

provenance guarantees (relational residue, structural chrome removal, translation-risk exoneration).

(three passes, 2-of-3 consensus, no-file-access sandbox, read-every- hit adjudication).

subject-matter domains and two verifier architectures, plus a graded refinement (DR5\\|\\|) across five DR6 variants.

seeing results. All 18 papers reproduce byte-identically.

- A methodology for building a real domain-specific corpus with
- A methodology for sandboxed subagent extraction and verification
- Two theorems (DR5, DR7) with an empirical triangulation across two
- Preregistered gates across every empirical run. Nothing tuned after

Not established:

actually works on real material. DCR2a tested only class-vs- proposition scoring, not the full DR pipeline.

would rank the T1 class first. Constructing sound g is DR7's open problem.

- Whether the DR-arc's deletion-repair *nominator* (from DR1–DR4)
- Whether class-aware nomination on the DCR corpus, with a *sound g*,
- Quantitative characterisation of the DR5\\|\\| gradient.

7. Natural follow-ups (all named in one paper or another)

Empirical extensions:

gradient. Systematically vary prompt specificity; measure overlap gap and stdev; fit a curve. specificity while keeping verifier fixed).

- **DR8** – quantitative characterisation of the semantic-access
- **DR6i** – target with degradable semantic depth (vary D's

Theorem extensions:

exists for a given D. Constructive criterion vs impossibility.

- **DR9** – formal characterisation of when a domain-general proxy

DCR arc extensions:

- **DCR2b** – multi-doc aggregation on DCR1e consensus.
- **DCR2c** – presupposition-inferring extractor as g itself.

8. Discipline summary

Nine principles that held across the arc:

prior modules.

matchers are drafted.

27%.

DR6, DR6c, DR6h all had "did not confirm the hypothesis" outcomes that turned into refinements.

1. Preregister before every run.
2. Every threshold as imported constant, not restated in the runner.
3. Every new experiment additive: new module, new runner, never edits
4. Sandboxed subagents forbid file access beyond the named document.
5. Consensus 2-of-3 over three independent passes.
6. Held-out validation sets committed at fixed SHA-256 BEFORE
7. Read every hit; DCR1's exemplar-only adjudication over-stated by
8. No post-hoc matcher tightening to escape a placebo firing.
9. Report the failure mode honestly when a run fails – DCR1f, DCR1e,

Appendix: reading order

Shortest path to the finding:

domain-general proxy.

1. **DR5** – the theorem.
2. **DCR1f** – the empirical wall on a real corpus with ground truth.
3. **DR6h** – the DR5\\|| graded picture from a target without a
4. **DR7** – why the grouping function doesn't escape.
5. **DR5b** – practical guidance for verifier designers.

Depth on the DCR corpus:

1. DCR1 (corpus + relational residue)
2. DCR1c (six-gate G0, T1 absent)
3. DCR1d (Newton positive control – historical, not instrumental)
4. DCR1e (presupposition extractor, recognizer gap)
5. DCR1f (target_v4, placebo fires)

Full DR6 empirical triangulation:

DR6 → DR6c → DR6d → DR6e → DR6f → DR6g → DR6h. Reading in order shows how each variant sharpened DR5 *through* DR5*.

All papers reproduce via each runner's own run_*.py. Prior verdicts stay byte-identical after every subsequent paper.